

# International AS and A-level

## Chemistry (9620)

### CH04 Organic 2 and Physical 2

### Report on the examination

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June 2024

## REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH04 JUNE 2024

In line with previous examination series, students generally performed better in the calculation questions and Physical Chemistry questions than in the Organic Chemistry questions. There were fewer students getting full marks for questions requiring naming or outlining mechanisms than usual.

### General points to note:

- at least two significant figures are required for all calculation questions
- check rounding of numbers if three significant figures are required
- learn definitions for AO1 marks
- make sure points plotted can be seen by examiner, crosses are easier to see than dots
- check to see whether a particular type of formula is requested eg displayed or skeletal
- arrows in mechanisms need to be drawn from a bond or from a lone pair of electrons
- learn the names of mechanisms

## QUESTION 01

Question 1 was answered well by most students. Around 90% students got full marks on 01.1, 01.2 and 01.3 and 75% got both marks for 01.4.

01.2 Despite being given the number of moles in the question, a few students used the wrong number of moles to calculate the mole fraction. Some just calculated the mole fraction and not the partial pressure.

01.3 Some students used square brackets in their equilibrium equation. Students do not need to use brackets at all to denote partial pressure but if they do these should be round brackets.

01.4 Most gave the first of the two sets of answers on the mark scheme. Some didn't mention the effect on  $K_p$  and some didn't realise that  $K_p$  was smaller for  $T_2$  than  $T_1$ . A few students mentioned effect on rate rather than equilibrium position.

## QUESTION 02

Question 2 was mathematical in the most part and around 80% students got full marks for finding the order of reaction (02.2) and calculating  $k$  (02.3). Fewer students were able to give a definition of the order of reaction (02.1) with only 50% getting this correct.

02.1 Many students did not mention the rate equation and some stated that order was 'how much concentration affects the rate'. This is a definition that needs to be memorised.

02.2 The most common error was to have the order with respect to B as 0

02.3 Most students answered this correctly. Those who lost marks tended to give incorrect units rather than value for  $k$ . Units have to be given in standard form. Some students only gave an answer to 1 significant figure or rounded incorrectly.

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## QUESTION 03

03.1 Most students were able to calculate the values for the table although some did not round the numbers correctly and so lost the mark. Fewer students were unable to find the rate constant from the natural log than were able to find  $1/T$ .

03.2 Around 75% got full marks for the graph. Some did not show where their points were. Students should draw points with a small cross so that it is easy to identify where they have plotted.

03.3 Around 70% were able to use their graph to find a gradient and calculate  $E_a$ . Some students did not use the graph and used one set of data from the table instead.

## QUESTION 04

Question 4 was less well answered than the previous three questions with less than 50% getting full marks for 04.2 and 04.3 (mechanism). It seemed that fewer were familiar with how to outline a mechanism than in previous series with some writing in words what happened in the mechanism. Students found 04.1 and 04.4 more accessible.

04.1 Most students were able to draw this structure although some had not realised that the skeletal formula was needed. A few students drew propanone. It was good to see that fewer students drew the H on the carbonyl C than in previous series.

04.2 Surprisingly only around 50% students identified this correctly. Some just wrote 'carboxylic acid' rather than naming the specific compound formed. A few thought that the propanal was reduced and wrote propan-1-ol or alcohol.

04.3 Some students named the mechanism as a reduction rather than nucleophilic addition. It seems that they were unclear about the difference between the type of reaction and the mechanism. Some used  $\text{BH}_4^-$  as the nucleophile rather than  $\text{H}^-$ . In some cases, arrows were incorrectly drawn from the -ve charge on the nucleophile and/or on the  $\text{O}^-$  in the intermediate. A few students did not draw an arrow from the  $\text{C}=\text{O}$  bond but from the C atom. There were a few students who wrote a description of what happens in the mechanism rather than representing it with curly arrows as required.

04.4 Some students referred to the absorption due to O-H bond being present. This was ignored as it does not show that the reaction is complete.

## QUESTION 05

05.1 Most students answered this correctly.

05.2 Not all students realised that they needed to give displayed formulas for the monomers so were awarded only one out of two marks available if they had identified the correct monomers but given a different type of formula. The O-H bond needs to be drawn when giving displayed formulas.

05.3 Very few students got both marks for this question as they failed to explain that it was the polar C-O bond in the chain that made hydrolysis possible.

05.4 Most students answered this well.

05.5 Less than 50% students got the structure correct, possibly because it was a cyclic structure.

## QUESTION 06

Over 70% were able to give the reagent and equation for 06.1 and equation for 06.2 but only 35% were able to name and outline the mechanism correctly in 06.3

06.1 Most students knew that the reagent was  $\text{AlCl}_3$  but some were unable to give the equation for the formation of the electrophile.

06.2 Most were able to give the equation. A few made balancing errors however.

06.3 Surprisingly few students got full marks in this question which was a standard mechanism that has been asked for frequently in previous series. Some students incorrectly named the mechanism as an addition reaction. Some did not put the positive charge on the N atom of the electrophile. A few didn't draw the intermediate correctly. The horseshoe needs to be centred on the carbon opposite the site of substitution and the positive charge should be within the horseshoe rather than on the C where the substitution is taking place.

## QUESTION 07

07.1 Many students seemed to be unfamiliar with atom economy with only 40% getting the mark. Some students did not mention the  $M_r$  and instead discussed the number of atoms being used to make aspirin.

07.2 Over 90% got this correct as there were many correct answers to this question.

07.3 Over half of students were able to give both marking points.

07.4 – 07.7 These were questions about Required Practical 10 and were generally answered well (60–70% correct) except 07.7 where only 40% got the mark. A number of students mentioned the aspirin melting rather than redissolving in 07.7.

## QUESTION 08

08.1 This was answered well by students, and they seemed more confident in giving uses of esters than other organic compounds we have asked about in previous series.

08.2 Over 85% were able to give the correct number of peaks.

08.3 Fewer than 60% students were able to give integration value with many giving a 3:1:3 ratio. Students should check that the sum of the values in their integration ratio matches the number of H atoms in the molecular formula (or is a multiple of it).

08.4 Most were able to name the splitting patterns correctly despite the requirement for correct terminology (not just 2 and 1 for example).

08.5 This was answered well with around two thirds of students getting all three marks.

## QUESTION 09

09.1 Less than 60% were able to give the reagent and mechanism despite this being a Unit 2 mechanism. The most common errors were bromine as a reagent and/or elimination or nucleophilic addition as the mechanism.

09.2 More students were able to give this reagent with over 70% getting it correct.

09.3 Only 60% got both marks for this question with many thinking the reagent was ammonia.

09.4 Around 70% were able to identify the compound needed.

09.5 Around a quarter of students got both marks for this question. Many students were not able to describe what a nucleophile and base are in terms of electron pair donation to an electron deficient C or a proton.

09.6 Less than 30% got this correct as it was an N-substituted amine. Some students thought it was a diamine and others thought it was an amide.

09.7 Over 50% got the quaternary structure correct which is higher than has been the case for similar questions in previous series.

## QUESTION 10

10.1 This was answered similarly to previous series with only 60% answering correctly. The main error seen were incorrect numbers or 'hydroxyl' rather than 'hydroxy'.

10.2 Surprisingly only just over 50% were able to get all three marks for describing optical isomers and how to identify them.

10.3 -10.6 These questions were answered well although some were only able to identify one error in the chromatography set up (10.5).

10.7 Around 55% students answered this correctly with many thinking the  $R_f$  value depended on the size of the chromatography paper.

10.8 Many students did not realise that the salts of the amino acids would be formed so only got one out of the two marks available.

10.9 Around two thirds of students were able to give both secondary structures.